Increasing interoperability of bioimage dataset resources

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NFDI 4
BIOIMAGE

Increasing interoperability of bioimage dataset resources

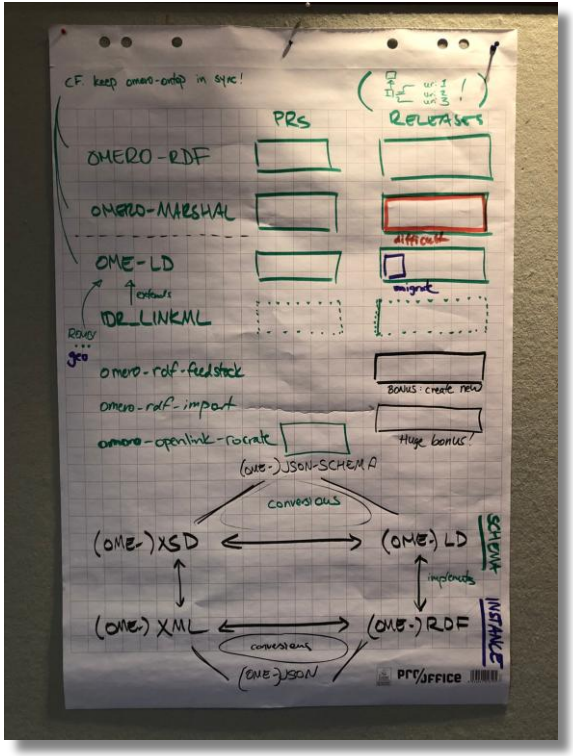
P1. Define ▶ P2. Extract ▶ P3.Benchmark

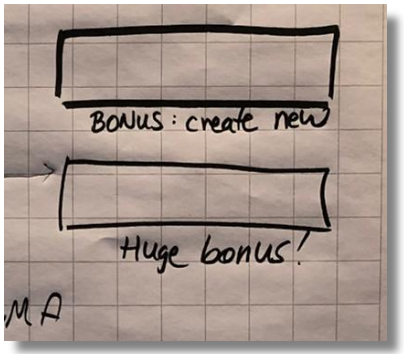
P1. “Correctness”, “Versions”, etc.

Zzzzzzz……



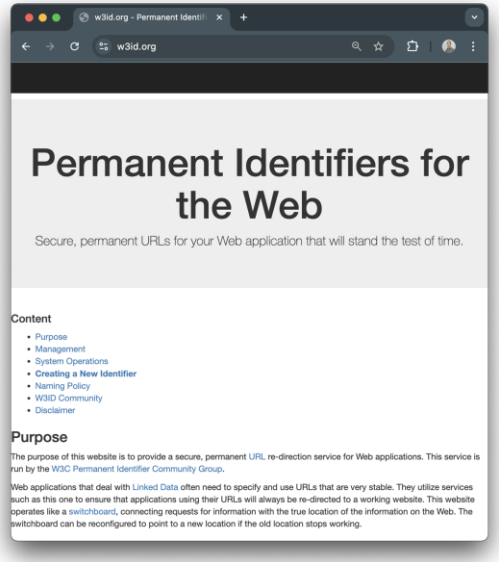
P1. Bonuses


CF. keep omero-ontap in sync!
PRs
RELEASES
OMERO-RDF
OMERO-MARSHAL
difficult
OME-LD
might
IDR_LINKML
Rewrite
in Go
omero-rdf-feedstock
BONUS: create new
omero-rdf_import
Huge bonus!
omero-openlink-rotrate
(OME-) JSON-SCHEMA
conversions
(OME-) XSD
(OME-) LD
SCHEMA
implements
INSTANCE
(OME-) XML
(OME-) RDF
conversions
(OME-) JSON
PROOFFICE

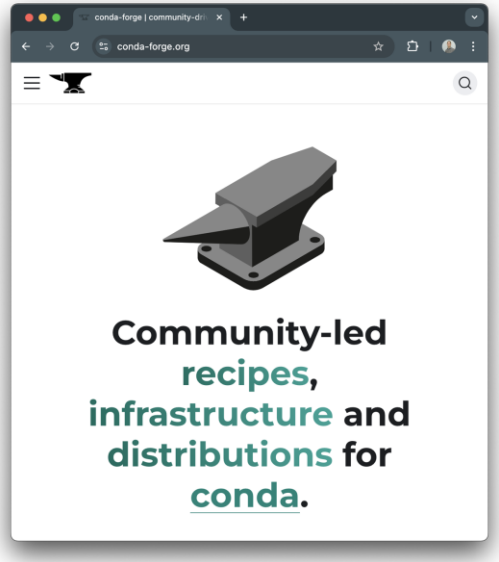

BONUS: create new
Huge bonus!

P1. Bonuses

Maintenance

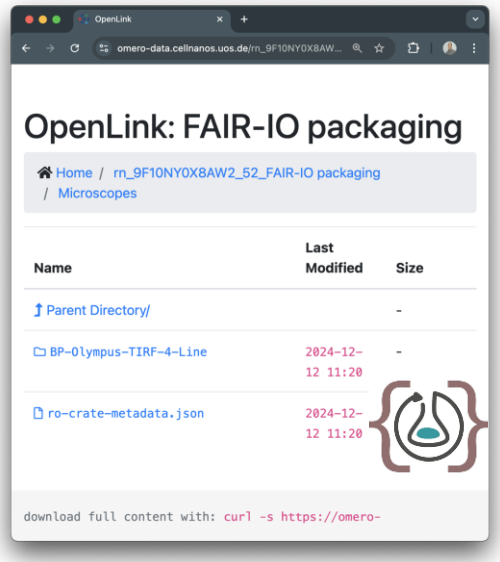


[w3id.org](w3id.org)



conda-forge.org

Usability



researchobject.org

P2.

Extract



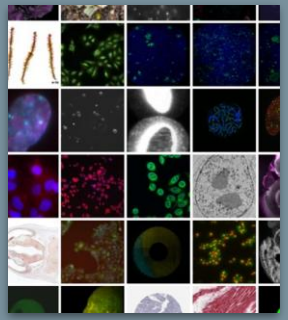


Photo by Albert Hyseni on Unsplash

Billions of Triples? **0.00**

P2. Use cases

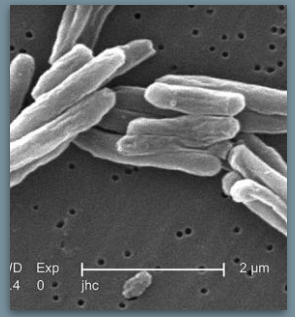
AI

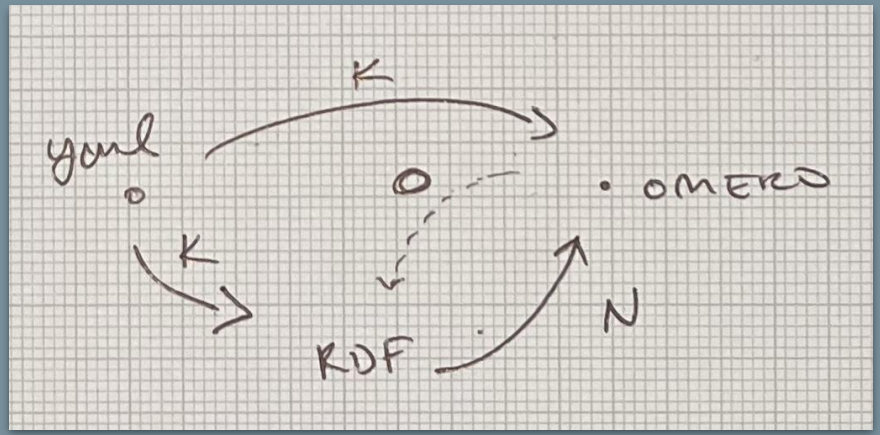


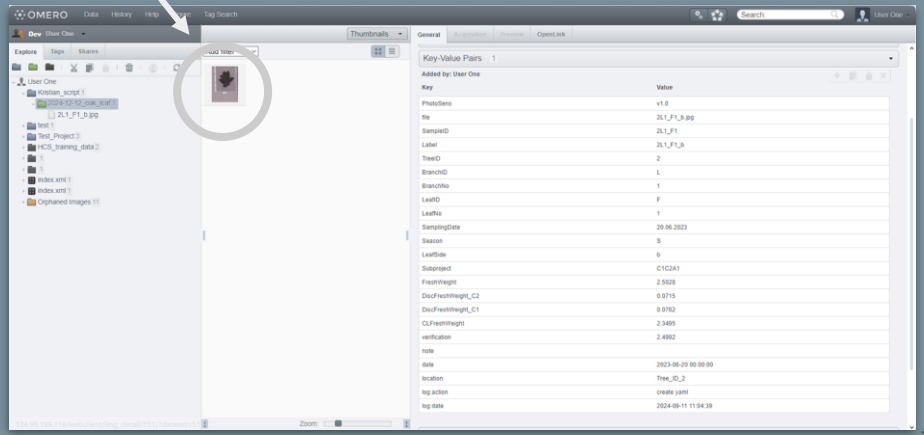
Oak

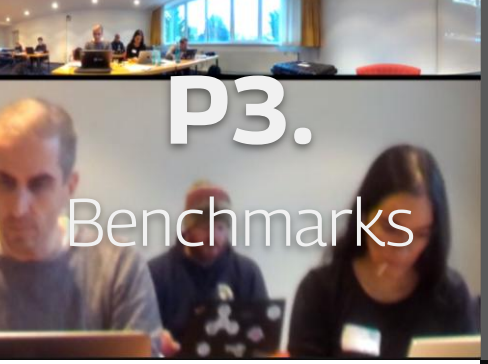


TB









P3.
Benchmarks

SPARQL Endpoint Drafting and Testing

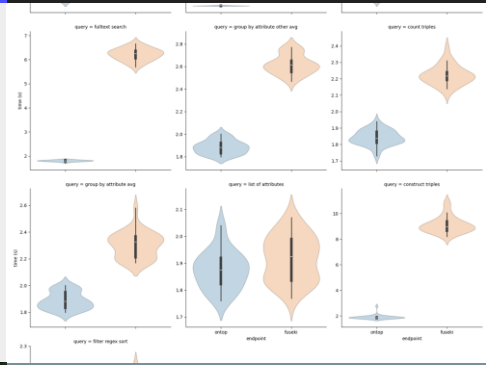
https://medium.com/wallscope/comparing-linked-data-triplestores-ebfac8c3ad4f

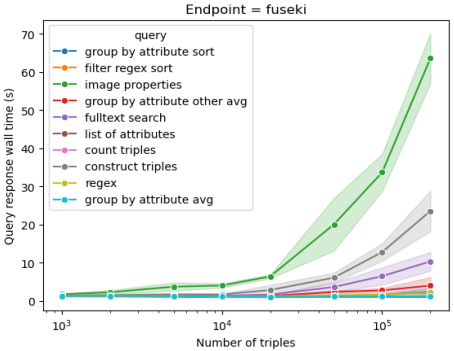
5 completed benchmark runs

3 **Endpoints**



10 **Queries**





3 triplestores benchmarked



500k triples

Test infrastructure

github.com/NFDI4BIOIMAGE/BioHackathon2024-P4



Westfälische Wilhelms-Universität Münster

Virtual KG

R2RML Mapping

Ontology

OMERO DB PostgreSQL

25k images + metadata

```
CONSTRUCT {?s ?p ?o} WHERE {?s ?p ?o}
```

500k triples

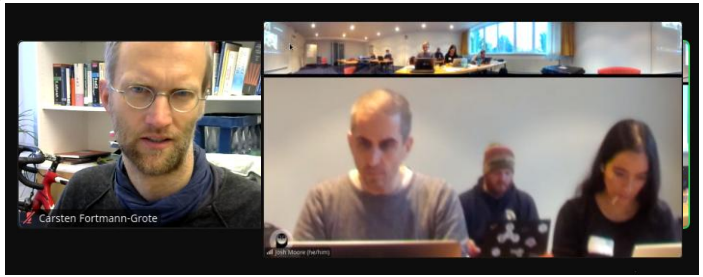
ontop

```
$> time rsparql –query 01.rq –service 128.176.233.7:8080/sparql
```

Materialized KG

Apache Jena Fuseki

virtuoso

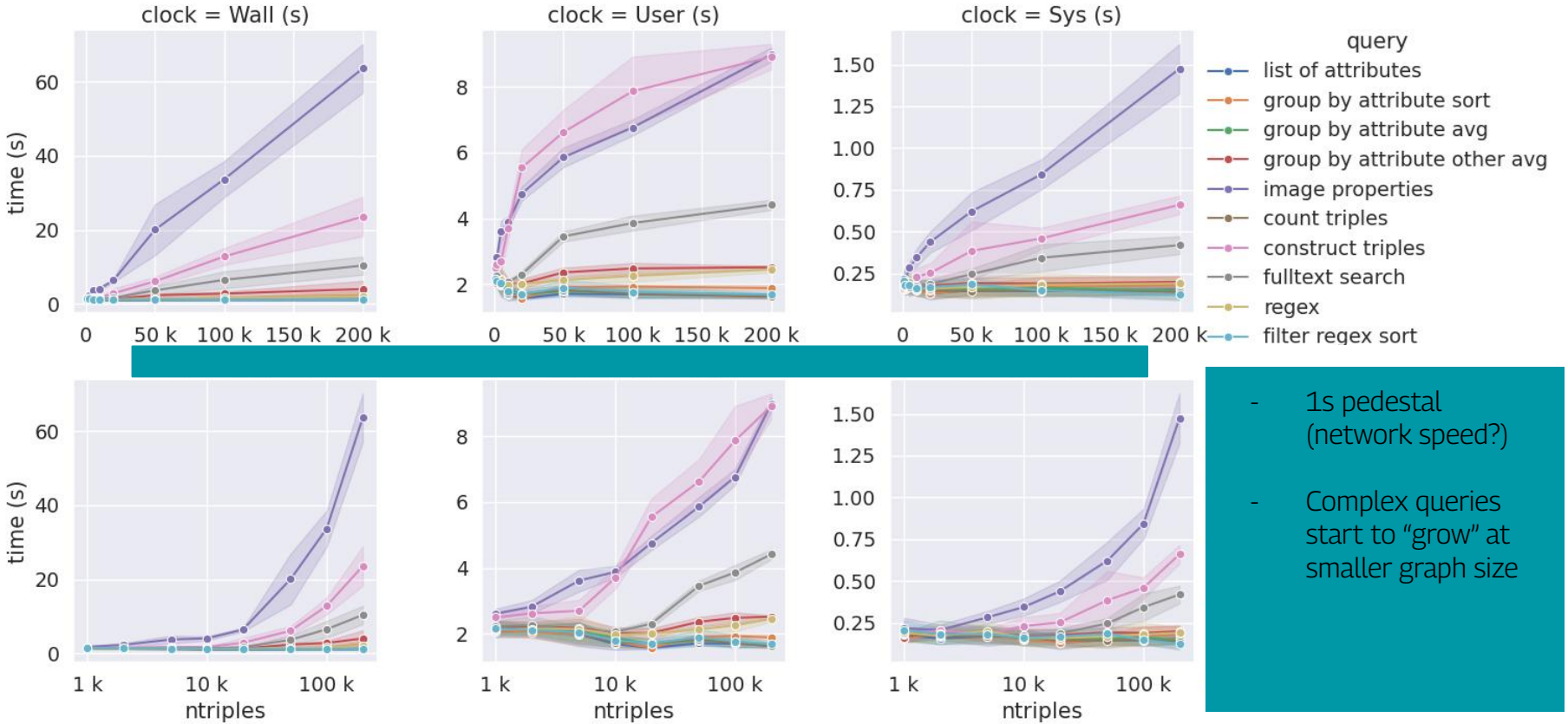


Material KG up to 50% faster than Virtual



Network? Resource sharing? Client load? Server load?

Query response time ∝ number of triples



- 1s pedestal (network speed?)
- Complex queries start to "grow" at smaller graph size

Network? Resource sharing? Client load? Server load?



10 **People**



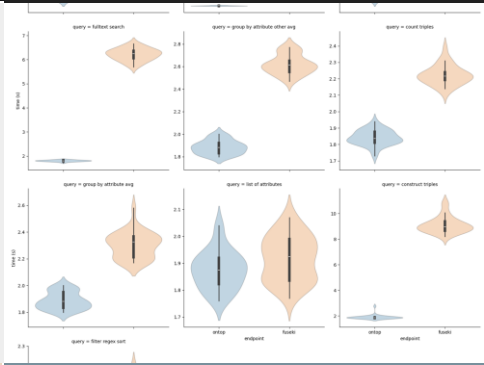
17 **Pulls**

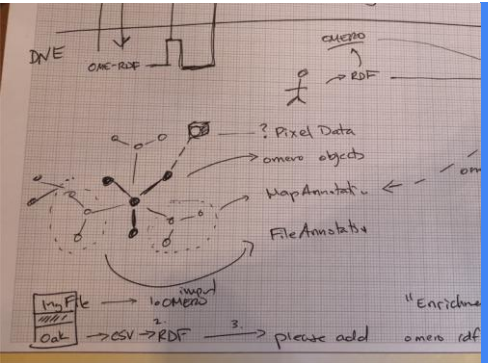
across 6 repositories

5 **First contributions!**



3 **New repositories**





14 **Issues**

(incl. a conda bug)



Oldest “TODO” fixed:

Nov 24 2016